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OBJECT HOLDING DEVICE, EXPOSURE DEVICE, OBJECT MOVING METHOD, AND OBJECT HOLDING SYSTEM

Abstract

An object holding device includes a holding unit that holds an object and is driven in a first direction and a second direction orthogonal to the first direction in a horizontal plane, and a pair of linear motors that apply thrusts in the first and second directions to the holding unit, each of the linear motors including a first unit including armature modules each including a magnetic core having two or more protruding portions, which protrude in the second direction, and a coil wound around the magnetic core, currents of the same phase flowing through the coil, and a second unit including a magnet module that includes permanent magnets arranged in the first direction while changing poles and is arranged between two adjacent protruding portions, wherein a part of each of the permanent magnets is accommodated in a space sandwiched between the two adjacent protruding portions.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation application of PCT/JP2023/043473, filed on Dec. 5, 2023, which claims the benefit of priority of the prior Japanese Patent Application No. 2022-201284, filed on Dec. 16, 2022, the entire contents of which are incorporated herein by reference.

FIELD

[0002] A certain aspect of embodiments described herein relates to an object holding device, an exposure device, an object moving method, and an object holding system.

BACKGROUND

[0003] In a lithography process for manufacturing electronic devices (microdevices) such as liquid crystal display devices and semiconductor devices (integrated circuits), a step-and-scan exposure device (so-called scanning stepper (also called a scanner)) is used which transfers the pattern formed on the mask onto a glass substrate or wafer (hereinafter collectively referred to as a “substrate”) by using an energy beam while synchronously moving the mask or reticle (hereinafter collectively referred to as a “mask”) and the substrate along a predetermined scanning direction (scan direction).

[0004] In exposure devices, the development of light sources represented by UV-LEDs (Ultraviolet-Light Emitting Diodes) has dramatically improved the illuminance of the illumination system, and the amount of exposure can be secured even when various stages, including mask stages, are driven at high speed. Therefore, there is an increasing demand for higher driving speeds of various stages including mask stages. The throughput is improved by increasing the driving speeds of various stages.

[0005] For example, a linear motor is used as a driving device to drive the mask stage in the scanning direction as disclosed in, for example, Japanese Patent Application Laid-Open No. 2017-15995 (Patent Document 1).

SUMMARY

[0006] In one aspect, there is provided an object holding device including: a holding unit that holds an object and is driven in a first direction, which is a scanning direction of the object, and a second direction orthogonal to the first direction in a horizontal plane; and a pair of linear motors that apply a thrust in the first direction and a thrust in the second direction to the holding unit, each of the pair of linear motors including a first unit and a second unit, the first unit including a plurality of armature modules, each of the armature modules including a magnetic core having two or more protruding portions, which protrude in the second direction, and a coil wound around the magnetic core, currents of the same phase flowing through the coil, and the second unit including a magnet module that includes a plurality of permanent magnets arranged in the first direction while changing poles and is arranged between two adjacent protruding portions of the two or more protruding portions, wherein at least a part of each of the permanent magnets is accommodated in a space sandwiched between the two adjacent protruding portions.

[0007] In a second aspect, there is provided an exposure device including: the above object holding

device; and a pattern formation device that forms a pattern that the object has onto an exposure target by an exposure operation of exposing the exposure target with an energy beam through the object held by the object holding device.

[0008] In a third aspect, there is provided an object moving method including: arranging a pair of first units so that a first direction is orthogonal to a scanning direction of an object, each of the pair of first units having a plurality of armature modules, each of the armature modules including a magnetic core having two or more protruding portions, which protrude in the first direction, and a coil wound around the magnetic core, currents of the same phase flowing through the coil; arranging a pair of second units each having a magnet module that includes a plurality of permanent magnets arranged in a second direction while changing poles in the second direction and is arranged between two adjacent protruding portions of the two or more protruding portions so that the second direction is parallel to the scanning direction and at least a part of each of the permanent magnets is accommodated in a space sandwiched between the two adjacent protruding portions; and applying a thrust in the scanning direction and a thrust in a direction orthogonal to the scanning direction in a horizontal plane to a holding unit that holds the object by a pair of linear motors each including the first unit and the second unit to move the object in the scanning direction and the direction orthogonal to the scanning direction.

[0009] In a fourth aspect, there is provided an object holding system including: a holding unit that holds an object and is driven in a first direction, which is a scanning direction of the object, and a second direction orthogonal to the first direction in a horizontal plane; a pair of linear motors that apply a thrust in the first direction and a thrust in the second direction to the holding unit, each of the pair of linear motors including a first unit and a second unit, the first unit having a plurality of armature modules, each of the armature modules including a magnetic core having two or more protruding portions, which protrude in the second direction, and a coil wound around the magnetic core, currents of the same phase flowing through the coil, the second unit having a magnet module that includes a plurality of permanent magnets arranged in the first direction while changing poles and is arranged between two adjacent protruding portions of the two or more protruding portions; and a control device configured to control the pair of linear motors, wherein at least a part of each of the permanent magnets is accommodated in a space sandwiched between the two adjacent protruding portions.

[0010] The configuration of the embodiments described below may be modified appropriately, and at least one or some of the components may be substituted for other components. Further, the constituent elements whose arrangement is not particularly limited are not limited to the arrangement disclosed in the embodiment, and can be arranged at positions where the functions can be achieved.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. 1 is a schematic view illustrating a configuration of an exposure device according to an embodiment;

[0012] FIG. 2A is a plan view of a mask stage device according to the embodiment, and FIG. 2B is a side view of the mask stage device according to the embodiment;

[0013] FIG. 3A is a perspective view illustrating a configuration of a linear motor, and FIG. 3B is a side view for describing an arrangement of a first unit and a second unit in the embodiment;

[0014] FIG. 4A and FIG. 4B are side views for describing a force generated between the first unit and the second unit, and FIG. 4C is a side view for describing a thrust force in the Y-axis direction applied to a stage body;

[0015] FIG. 5 is a block diagram illustrating a configuration of a mask stage control device that

controls the driving of the first linear motor and the second linear motor; and [0016] FIG. 6A is a plan view of the mask stage device according to a variation, and FIG. 6B is a side view of the mask stage device according to the variation.

DESCRIPTION OF EMBODIMENTS

[0017] As described above, for example, a linear motor is used as a driving device for driving the mask stage in the scanning direction. The mask stage is configured as a three degree-of-freedom (DOF) plane stage with the X-axis, Y-axis, and θz -axis having degrees of freedom on a plane in order to synchronize with the substrate stage.

[0018] When a linear motor with a core, which is a highly efficient linear motor having a large flux density, is used to meet the demand for higher driving speed of the mask stage, it is difficult to implement the linear motor to a 3-DOF plane stage such as the mask stage because a strong magnetic attraction force is generated between the core portion and the permanent magnets in a typical linear motor with a core. In the present embodiment, a linear motor with a core having a large magnetic flux density and high efficiency is implemented to the mask stage to achieve higher driving speed of the mask stage and position control of the mask stage in the X-axis direction, the Y-axis direction, and the θz direction.

[0019] An exposure device **10** according to an embodiment will be described with reference to FIG. 1 to FIG. 5.

(Configuration of Exposure Device)

[0020] FIG. 1 is a view schematically illustrating a configuration of the exposure device **10** according to an embodiment.

[0021] The exposure device **10** is a scanning stepper (scanner) that transfers the pattern formed on a mask MSK onto a glass substrate (hereinafter referred to as a “substrate”) P by driving the mask MSK and the substrate P in the same direction at the same speed with respect to a projection optical system PL. The substrate P is, for example, a rectangular glass substrate used for a liquid crystal display device (flat panel display), and at least one side or diagonal of the substrate P is equal to or longer than 500 mm.

[0022] In the following description, the direction (scanning direction) in which the mask MSK and the substrate P are driven during scanning exposure is defined as the X-axis direction, the direction orthogonal to the X-axis direction in the horizontal plane is defined as the Y-axis direction, the direction orthogonal to the X-axis and Y-axis is defined as the Z-axis direction, and the rotational (tilt) directions about the X-axis, Y-axis, and Z-axis are defined as the θx , θy , and θz directions, respectively. Further, the position in the X-axis direction, the position in the Y-axis direction, and the position in the θz direction of a stage body **60** of a mask stage device MST to be described later will be sometimes referred to as the X position, the Y position, and the θz position, respectively.

[0023] The exposure device **10** includes an illumination system IOP, the mask stage device MST that holds the mask MSK, the projection optical system PL, a body **70** that supports these components, a substrate stage PST that holds a substrate P, and a control device **600**. The control device **600** includes a mask stage control device **400** that controls the mask stage device MST, and a substrate stage control device **500** that controls the substrate stage PST.

[0024] The body **70** includes a base (vibration isolator) **71**, columns **72A** and **72B**, and an optical surface plate **73**. The base (vibration isolator) **71** is placed on the floor F, and supports the columns **72A** and **72B**, and the like while isolating vibration from the floor F. The columns **72A** and **72B** each have a frame shape, and the column **72B** is disposed inside the column **72A**. The optical surface plate **73** has a flat plate shape and is fixed to the ceiling of the column **72A**.

[0025] The illumination system IOP is arranged above the body **70**. The illumination system IOP irradiates the mask MSK with the illumination light IL.

[0026] The mask stage device MST includes the stage body **60**, on which the mask MSK having a pattern surface (the lower surface in FIG. 1) on which a circuit pattern is formed is fixed by, for example, vacuum suction (or electrostatic adsorption). The mask stage device MST is driven by a

pair of linear motors **100** (described later) in the scanning direction (X-axis direction) with a predetermined stroke, and is finely driven in non-scanning directions (Y-axis direction and θ_z direction). The configuration of the mask stage device MST will be described later in detail.

[0027] The projection optical system PL is supported by the optical surface plate **73** below (at the $-Z$ side of) the mask stage device MST. The projection optical system PL forms, for example, a rectangular image field whose longitudinal direction is the Y-axis direction. The projection area of the projection optical system PL may be referred to as an exposure area.

[0028] When the illumination area on the mask MSK is illuminated with the illumination light IL from the illumination system IOP, a projection image (partial erect image) of the circuit pattern of the mask MSK in the illumination area is formed on the irradiation area (exposure area (conjugate to the illumination area)) on the substrate P arranged on the image plane side of the projection optical system PL through the projection optical system PL by the illumination light IL transmitted through the mask MSK. Here, a resist (sensitive agent) is applied to the surface of the substrate P. The mask stage device MST and the substrate stage PST are synchronously driven, that is, the mask MSK is driven in the scanning direction (X-axis direction) with respect to the illumination area (illumination light IL) and the substrate P is driven in the same scanning direction with respect to the exposure area (illumination light IL), whereby the substrate P is exposed and the pattern of the mask MSK is transferred onto the substrate P.

[0029] The substrate stage PST is arranged on the base (vibration isolator) **71** below (at the $-Z$ side of) the projection optical system PL. The substrate P is held on the substrate stage PST via a substrate holder (not illustrated).

[0030] The position information (including rotation information (the yawing amount (rotation amount θ_z in the θ_z direction), the pitching amount (rotation amount θ_x in the θ_x direction), the rolling amount (rotation amount θ_y in the θ_y direction)) of the substrate stage PST in the XY plane is measured by the interferometer system. The interferometer system measures the position of the substrate stage PST by emitting a measurement beam from the optical surface plate **73** onto a moving mirror (or a mirror-finished reflective surface (not illustrated)) on the end portion of the substrate stage PST, and receiving the reflected light from the moving mirror. The measurement results are supplied to the substrate stage control device **500**, and the substrate stage control device **500** drives the substrate stage PST in accordance with the measurement results of the interferometer system.

[0031] The exposure device **10** performs alignment measurement (for example, EGA or the like) prior to exposure, and performs the exposure of the substrate P using the results of the measurement according to the following procedure. First, the mask stage device MST and the substrate stage PST are synchronously driven in the X-axis direction in accordance with instructions from the mask stage control device **400** and the substrate stage control device **500**. Then, scanning exposure is performed on the first shot area on the substrate P. When the scanning exposure for the first shot area is completed, the substrate stage control device **500** moves (steps) the substrate stage PST to a position corresponding to the second shot area. Then, scanning exposure is performed on the second shot area. The substrate stage control device **500** repeats stepping between shot areas of the substrate P and scanning exposure for the shot areas in the same manner. As a result, the pattern of the mask MSK is transferred to all the shot areas on the substrate P.

(Configuration of Mask Stage Device MST)

[0032] Next, the configuration of the mask stage device MST in the present embodiment will be described. FIG. 2A is a plan view of the mask stage device MST, and FIG. 2B is a side view of the mask stage device MST. In FIG. 2B, a second unit **300** described later is illustrated in a cross section including a permanent magnet **301** described later. In FIG. 2B, some components are not illustrated.

[0033] As illustrated in FIG. 2A and FIG. 2B, the mask stage device MST includes a pair of X

beams **61**, the stage body **60** that holds the mask MSK, a pair of linear motors **100** that apply thrust to the stage body **60**, and the like.

[0034] The pair of X beams **61** are fixed to, for example, the column **72B**. The pair of X beams **61** are members extending in the X-axis direction, and are arranged in parallel to each other, separated in the Y-axis direction.

[0035] The stage body **60** is formed of a plate-like member that is rectangular in a plan view, and an opening **60a** in the form of a rectangular long hole whose longitudinal direction is in the X-axis direction is formed in the center portion of the stage body **60**. The mask MSK is inserted into the opening **60a**. A plurality of (for example, five) holding members (not illustrated) including suction pads for sucking and holding the mask MSK from below are attached to each of the wall surfaces on the +Y side and the -Y side of the wall surfaces defining the opening **60a** at predetermined intervals in the X-axis direction.

[0036] Further, air bearings **62**, which are a type of static gas bearing, are attached to the lower surface of the stage body **60**, for example, in the vicinities of the four corners of the lower surface of the stage body **60** (see FIG. 2B). The stage body **60** floats in a non-contact manner via a minute clearance over a surface plate **65** supported by the column **72B**, by blowing pressurized gas from, for example, the four air bearings **62**.

[0037] Further, as illustrated in FIG. 2A, a pair of X moving mirrors **63X** having a reflecting surface orthogonal to the X-axis are attached to the side surface on the -X side of the stage body **60** at a predetermined interval in the Y-axis direction. Further, a Y moving mirror **63Y** (bar mirror) having a reflecting surface orthogonal to the Y-axis is attached to the side surface on the -Y side of the stage body **60**.

[0038] The positional information of the stage body **60** (that is, the mask MSK) in the XY plane is constantly detected at a resolution of, for example, about 0.5 to 1 nm by a laser interferometer system (hereinafter referred to as a mask interferometer system) including a pair of X laser interferometers **64X** corresponding to the pair of X moving mirrors **63X**, respectively, and Y laser interferometers **64Y** corresponding to the Y moving mirror **63Y**. The position information of the stage body **60** in the θz direction is obtained based on the outputs of the pair of X laser interferometers **64X**.

[0039] The stage body **60** is driven in the X-axis direction, the Y-axis direction, and the θz direction by a pair of linear motors **100**. The pair of linear motors **100** are opposed to each other in the Y-axis direction with the stage body **60** interposed therebetween. To be more specific, the pair of linear motors **100** includes a first linear motor **100a** arranged at the +Y side of the stage body **60** and a second linear motor **100b** arranged at the -Y side of the stage body **60**.

(Configuration of Linear Motor **100**)

[0040] Next, a configuration of the linear motor **100** will be described. FIG. 3A is a perspective view illustrating the configuration of the linear motor **100**. As illustrated in FIG. 3A, the linear motor **100** includes a first unit **200** and the second unit **300**.

[0041] The first unit **200** includes a plurality of armature module sets **211** arranged in the X-axis direction. Each armature module set **211** includes armature modules **210U**, **210V**, and **210W**. In the following description, the armature modules **210U**, **210V**, and **210W** are referred to as armature modules **210** when no particular distinction is necessary. The armature modules **210U**, **210V**, and **210W** are housed in a housing **250** (see FIG. 2A) having an E-shaped cross section, for example.

[0042] Each armature module **210** includes a magnetic core **201** having three protruding portions **201a** that protrude in the Y-axis direction, and a coil **203** wound around the magnetic core **201**. A U-phase voltage is applied to the coil **203** of the armature module **210U**, a V-phase voltage is applied to the coil **203** of the armature module **210V**, and a W-phase voltage is applied to the coil **203** of the armature module **210W**. That is, currents in the same phase (U phase) flow through the coils **203** of the armature modules **210U**, respectively. Further, currents in the same phase (V phase) flow through the coils **203** of the armature modules **210V**, respectively. Further, currents in

the same phase (W phase) flow through the coils **203** of the armature modules **210W**, respectively. [0043] In each of the armature modules **210U**, **210V**, and **210W**, the winding directions of the coils **203** through which currents in the same phases flow are adjusted so that the magnetic flux closed loops are formed so that the polarities of the electromagnets of the respective protruding portions **201a** are different from each other. For example, as illustrated in the armature module **210U** of FIG. 3A, the winding direction of each coil **203** is adjusted so that the magnetic flux closed loops formed at a certain point of time are the magnetic flux closed loops A1 and A2.

[0044] The second unit **300** includes a magnet module **310** including a plurality of permanent magnets **301**. In the present embodiment, the second unit **300** includes two magnet modules **310**. In each magnet module **310**, the plurality of permanent magnets **301** are arranged while changing the poles in the X-axis direction.

[0045] FIG. 3B is a side view for describing the arrangement of the first unit **200** and the second unit **300** in the present embodiment. In FIG. 3B, a cross section including the permanent magnet **301** is illustrated for the second unit **300**. As illustrated in FIG. 3B, each magnet module **310** is disposed between two adjacent protruding portions **201a** of the magnetic core **201** of the armature module **210**. In FIG. 3A and FIG. 3B, “N” and “S” represent the N pole and the S pole of the permanent magnet **301**, respectively.

[0046] The coil **203** of at least one armature module **210** is supplied with a current having a different phase from that of the current supplied to the coil **203** of another armature module **210** so that a thrust in the X-axis direction is generated by the attraction and repulsion forces between the poles of the electromagnets formed at the ends of the protruding portions **201a** of each armature module **210** and the corresponding permanent magnets **301**. As the linear motor **100** having the above-described configuration, for example, a linear motor disclosed in Japanese Patent No. 5956993 can be adopted.

[0047] In the present embodiment, as illustrated in FIG. 2B and FIG. 3B, the first unit **200** is fixed to the X beam **61**, and the second unit **300** is fixed to the stage body **60**. In the present embodiment, the first unit **200** and the second unit **300** are arranged so that a part of each of the plurality of permanent magnets **301** included in the magnet module **310** is accommodated in the space SP1 sandwiched between two adjacent protruding portions **201a** of the magnetic core **201**, and the other part of each of the plurality of permanent magnets **301** is exposed from the space SP1. As a result, as illustrated in FIG. 3B, a magnetic attraction force MAF is generated between the first unit **200** and the second unit **300** in the Y-axis direction. It is sufficient if at least a part of each of the plurality of permanent magnets **301** is accommodated in the space SP1 sandwiched between two adjacent protruding portions **201a** of the magnetic core **201**.

[0048] The magnitude of the magnetic attraction force MAF generated between the first unit **200** and the second unit **300** varies depending on the area of the portion of the permanent magnet **301** that is accommodated in the space SP1 sandwiched between the two adjacent protruding portions **201a** of the magnetic core **201**. In the linear motor **100** according to the present embodiment, when the entire permanent magnet **301** is accommodated in the space SP1, the magnetic attraction force MAF is the smallest. Since a large magnetic attraction force MAF between the first unit **200** and the second unit **300** affects the driving of the linear motor **100**, conventionally, when the linear motor **100** according to the present embodiment is used, the first unit **200** and the second unit **300** are arranged so that the entire permanent magnet **301** is accommodated in the space SP1.

[0049] The inventor has found that the thrust in the Y-axis direction applied to the stage body **60** by the first linear motor **100a** and the thrust in the Y-axis direction applied to the stage body **60** by the second linear motor **100b** can be changed by the d-axis current by generating the magnetic attraction force MAF having an appropriate magnitude between the first unit **200** and the second unit **300**.

[0050] Therefore, in the present embodiment, the magnetic attraction force MAF is intentionally generated between the first unit **200** and the second unit **300**, and the d-axis current applied to the

coil **203** is changed, whereby the thrust in the Y-axis direction applied to the stage body **60** is changed in each of the first linear motor **100a** and the second linear motor **100b** by the forces caused by the increase and decrease of the magnetic fluxes due to the d-axis current. This allows the thrust of the stage body **60** in the Y-axis direction to be adjusted, and allows the position of the stage body **60** in the Y-axis direction to be adjusted.

[0051] The control of the position of the stage body **60** in the Y-axis direction by the control of the d-axis current will be described in detail. FIG. **4A** and FIG. **4B** are side views for describing the forces generated between the first unit **200** and the second unit **300**, and FIG. **4C** is a side view for describing the thrust in the Y-axis direction applied to the stage body **60**. FIG. **4A** to FIG. **4C** illustrate a cross section including the permanent magnet **301**, for the second unit **300**.

[0052] The left side of FIG. **4A** illustrates a state before the d-axis current is supplied to the coil **203**. The magnetic flux lines illustrated in FIG. **4A** are due to the q-axis current. As illustrated in FIG. **4A**, in a state where the d-axis current is not supplied to the coil **203** (d-axis current=0), a magnetic attraction force MAF is generated between the first unit **200** and the second unit **300**. When a positive d-axis current is supplied to the coil **203** in the state illustrated in FIG. **4A**, a force GF1 is generated between the first unit **200** and the second unit **300** due to an increase or decrease in the magnetic flux, as illustrated in the center of FIG. **4A**. The force GF1 is the force opposite in direction to the magnetic attraction force MAF.

[0053] Although the magnetic attraction force MAF is also generated between the first unit **200** and the second unit **300**, the magnetic attraction force MAF is weakened by the force GF1, and therefore, the total force TF of the force GF1 and the magnetic attraction force MAF becomes smaller than the magnetic attraction force MAF. As a result, the total force TF (<the magnetic attraction force MAF) of the force GF1 and the magnetic attraction force MAF is the force generated between the first unit **200** and the second unit **300**, as illustrated in the right side of FIG. **4A**.

[0054] The left side of FIG. **4B** illustrates a state before the d-axis current is supplied to the coil **203**, as in FIG. **4A**. In the state of FIG. **4B**, when a negative d-axis current is supplied to the coil **203**, as illustrated in the center of FIG. **4B**, a force GF2 is generated between the first unit **200** and the second unit **300** due to an increase or decrease in the magnetic flux. The force GF2 is the force in the same direction as the magnetic attraction force MAF.

[0055] The magnetic attraction force MAF is also generated between the first unit **200** and the second unit **300**, and the magnetic attraction force MAF is strengthened by the force GF2, so that the total force TF of the force GF2 and the magnetic attraction force MAF becomes larger than the magnetic attraction force MAF. As a result, as illustrated in the right side of FIG. **4B**, the total force TF (>the magnetic attraction force MAF) of the force GF2 and the magnetic attraction force MAF is generated between the first unit **200** and the second unit **300**.

[0056] The first unit **200** and the second unit **300** of the first linear motor **100a** and the first unit **200** and the second unit **300** of the second linear motor **100b** are arranged so that the respective magnetic attraction forces MAF become magnetic attraction forces of appropriate magnitudes from the viewpoint of the position control of the stage body **60** in the Y-axis direction. Further, the magnetic attraction force MAF generated between the first unit **200** and the second unit **300** of the first linear motor **100a** and the magnetic attraction force MAF generated between the first unit **200** and the second unit **300** of the second linear motor **100b** may be the same or different.

[0057] In the present embodiment, as illustrated in FIG. **4C**, the first linear motor **100a** is arranged at the +Y side of the stage body **60**, and the second linear motor **100b** is arranged at the -Y side of the stage body **60**. Assume that the mask stage control device **400** supplies a positive d-axis current to the second linear motor **100b** and supplies a negative d-axis current to the first linear motor **100a**. In this case, as illustrated in FIG. **4C**, a total force TF1 in the +Y direction is applied to the stage body **60** by the first linear motor **100a**, and a total force TF2 in the -Y direction is applied to the stage body **60** by the second linear motor **100b**. That is, the direction of the thrust in the Y-axis

direction applied to the stage body **60** by the first linear motor **100a** is opposite to the direction of the thrust in the Y-axis direction applied to the stage body **60** by the second linear motor **100b**.
[0058] As illustrated in FIG. 4C, since the total force TF1 in the +Y direction becomes larger than the total force TF2 in the -Y direction, a thrust THF in the +Y direction is applied to the stage body **60** as a whole, and the stage body **60** moves in the +Y direction. In this manner, the position of the stage body **60** in the Y-axis direction can be controlled by controlling the d-axis current supplied to the coil **203** of the first linear motor **100a** and the d-axis current supplied to the coil **203** of the second linear motor **100b**.

[0059] The d-axis current supplied to the coil **203** of the first linear motor **100a** and the d-axis current supplied to the coil **203** of the second linear motor **100b** can be determined by calculating the thrust applied to the stage body **60** in the Y-axis direction and distributing the thrust to the first linear motor **100a** and the second linear motor **100b**.

[0060] Specifically, the d-axis current supplied to the coil **203** of the first linear motor **100a** and the d-axis current supplied to the coil **203** of the second linear motor **100b** are determined so that the total force in the linear motor **100** located in the direction (+Y direction or -Y direction) in which the stage body **60** is to be moved (the total force in the direction in which the stage body **60** is to be moved) is larger than the total force in the other linear motor **100** (the total force in the direction opposite to the direction in which the stage body **60** is to be moved).

[0061] Here, when the d-axis currents in opposite directions are applied to the coil **203** of the first linear motor **100a** and the coil **203** of the second linear motor **100b**, respectively, the largest thrust can be generated in the Y-axis direction.

[0062] It is not necessary to apply the d-axis currents to the coils **203** of both the first linear motor **100a** and the second linear motor **100b**, and it is sufficient to apply the d-axis current to the coil **203** of at least one of the first linear motor **100a** or the second linear motor **100b**. That is, if the total force (total force in the direction in which the stage body **60** is to be moved) in the linear motor **100** positioned in the direction in which the stage body **60** is to be moved (+Y direction or -Y direction) is larger than the total force (total force in the direction opposite to the direction in which the stage body **60** is to be moved) in the other linear motor **100**, the stage body **60** can be moved in the desired direction.

[0063] FIG. 5 is a block diagram illustrating an example of the configuration of the mask stage control device **400** that controls the driving of the first linear motor **100a** and the second linear motor **100b**.

[0064] The mask stage control device **400** includes adder-subtractor circuits **401** to **403**, an X position control unit (X pos. control unit) **411**, a θz position control unit (θz pos. control unit) **412**, a Y position control unit (Y pos. control unit) **413**, an X-axis thrust calculation unit (X-axis thrust calc. unit) **460**, a Y-axis thrust calculation unit (Y-axis thrust calc. Unit) **470**, a magnetic attraction force calculation unit (MAF calc. unit) **480**, a q-axis current command value calculation unit (q-axis current cmd value calc. unit) **420**, a d-axis current command value calculation unit (d-axis current cmd value calc. unit) **430**, a first current-vector control unit (1st current-vector control unit) **440a**, a second current-vector control unit (2nd current-vector control unit) **440b**, a first motor amplifier (1 st motor amplifier) **450a**, and a second motor amplifier (2nd motor amplifier) **450b**.

[0065] The adder-subtractor circuit **401** calculates a deviation (X pos. deviation) between a target value of the X position of the stage body **60** input from the external device (X pos. target value) and the X position of the stage body **60** detected by the mask interferometer system (X actual pos.).

[0066] The adder-subtractor circuit **402** obtains a deviation (θz pos. deviation) between a target value of the θz position of the stage body **60** input from the external device (θz pos. target value) and the θz position of the stage body **60** detected by the mask interferometer system (θz actual pos.).

[0067] The adder-subtractor circuit **403** calculates a deviation (Y pos. deviation) between a target value of the Y position of the stage body **60** input from the external device (Y pos. target value) and

the Y position of the stage body **60** detected by the mask interferometer system (Y actual pos.).

[0068] The X-position control unit **411** calculates a command value of thrust in the X-axis direction (X-axis thrust cmd value) from the deviation of the X position calculated by the adder-subtractor circuit **401**.

[0069] The θz position control unit **412** calculates a command value of thrust in the θz direction (θz -direction thrust cmd value) from the deviation of the θz position calculated by the adder-subtractor circuit **402** (θz pos. deviation).

[0070] The X-axis thrust calculation unit **460** calculates a command value of the thrust in the X-axis direction of the first linear motor **100a** (1st linear motor X-axis thrust cmd value) and a command value of the thrust in the X-axis direction of the second linear motor **100b** (2nd linear motor X-axis thrust cmd value) based on the command value of the thrust in the X-axis direction calculated by the X position control unit **411** and the command value of the thrust in the θz direction calculated by the θz position control unit **412**.

[0071] The q-axis current command value calculation unit **420** calculates a command value of the q-axis current for the first linear motor **100a** (1st linear motor q-axis current cmd value) and a command value of the q-axis current for the second linear motor **100b** (2nd linear motor q-axis current cmd value) based on the X-axis thrust command value of the first linear motor **100a** and the X-axis thrust command value of the second linear motor **100b** calculated by the X-axis thrust calculation unit **460**.

[0072] On the other hand, the Y position control unit **413** calculates a command value of the thrust in the Y-axis direction (Y-axis thrust cmd value) from the deviation of the Y position (Y pos. deviation) calculated by the adder-subtractor circuit **403**.

[0073] The Y-axis thrust calculation unit **470** calculates a command value of the thrust in the Y-axis direction of the first linear motor **100a** (1st linear motor Y-axis thrust cmd value) and a command value of the thrust in the Y-axis direction of the second linear motor **100b** (2nd linear motor Y-axis thrust cmd value) based on the command value of the thrust in the Y-axis direction calculated by the Y-position control unit **413**.

[0074] The magnetic attraction force calculation unit **480** calculates the magnetic attraction force of the first linear motor **100a** (1st linear motor MAF) and the magnetic attraction force of the second linear motor **100b** (2nd linear motor MAF) from the actual position of the stage body **60**. The magnetic attraction force MAF varies depending on the extent to which each permanent magnet **301** is accommodated in the space SP1 between the adjacent protruding portions **201a** of the magnetic core **201**. The magnetic attraction force MAF also varies when the positional relationship between the first unit **200** and the second unit **300** slightly changes due to the driving of the stage body **60** in the X-axis direction. Therefore, in the present embodiment, the magnetic attraction force calculation unit **480** calculates the positional relationship between the magnetic cores **201** and the permanent magnets **301** based on the X actual position, the Y actual position, and the θz actual position of the stage body **60**, and calculates the magnetic attraction force of the first linear motor **100a** and the magnetic attraction force of the second linear motor **100b** from the calculated positional relationship.

[0075] The d-axis current command value calculation unit **430** calculates a d-axis current command value for the first linear motor **100a** (1st linear motor d-axis current cmd value) based on the Y-axis thrust command value of the first linear motor **100a** calculated by the Y-axis thrust calculation unit **470** and the magnetic attraction force of the first linear motor **100a** calculated by the magnetic attraction force calculation unit **480**. The d-axis current command value calculation unit **430** calculates a d-axis current command value for the second linear motor **100b** (2nd linear motor d-axis current cmd value) based on the Y-axis thrust command value of the second linear motor **100b** calculated by the Y-axis thrust calculation unit **470** and the magnetic attraction force of the second linear motor **100b** calculated by the magnetic attraction force calculation unit **480**.

[0076] The first current-vector control unit **440a** calculates command values of the U-phase, V-

phase, and W-phase voltages to be applied to the first linear motor **100a** (1st UVW-phase voltage cmd values) based on the q-axis current command value for the first linear motor **100a** calculated by the q-axis current command value calculation unit **420**, the d-axis current command value for the first linear motor **100a** calculated by the d-axis current command value calculation unit **430**, and the actual q-axis current and the actual d-axis current of the first linear motor **100a** detected by a detection unit (not illustrated), and outputs the calculated command values to the first motor amplifier **450a**.

[0077] The second current-vector control unit **440b** calculates command values of the U-phase, V-phase, and W-phase voltages to be applied to the second linear motor **100b** (2nd UVW-phase voltage cmd values) based on the q-axis current command value for the second linear motor **100b** calculated by the q-axis current command value calculation unit **420**, the d-axis current command value for the second linear motor **100b** calculated by the d-axis current command value calculation unit **430**, and the actual q-axis current and the actual d-axis current of the second linear motor detected by the detection unit (not illustrated), and outputs the calculated command values to the second motor amplifier **450b**.

[0078] The first motor amplifier **450a** applies voltages of the U-phase, the V-phase, and the W-phase (1st UVW-phase voltages) to the armature modules **210U**, **210V**, and **210W** of the first linear motor **100a**, respectively, in accordance with the voltage command values of the U-phase, the V-phase, and the W-phase input from the first current-vector control unit **440a**.

[0079] The second motor amplifier **450b** applies voltages of the U-phase, the V-phase, and the W-phase (2nd UVW-phase voltages) to the armature modules **210U**, **210V**, and **210W** of the second linear motor **100b**, respectively, in accordance with the voltage command values of the U-phase, the V-phase, and the W-phase input from the second-current vector control unit **440b**.

[0080] In this manner, the X position and the θ_z position of the stage body **60** can be adjusted by controlling the thrust of the first linear motor **100a** in the X-axis direction and the thrust of the second linear motor **100b** in the X-axis direction by the q-axis current, and the Y position of the stage body **60** can be adjusted by controlling the thrust of the first linear motor **100a** in the Y-axis direction and the thrust of the second linear motor **100b** in the Y-axis direction by the d-axis current.

[0081] As described in detail above, according to the present embodiment, the mask stage device MST includes the stage body **60** that holds the mask MSK and is driven in the X-axis direction, which is the scanning direction of the mask MSK, and the Y-axis direction, which is orthogonal to the X-axis direction in the horizontal plane, and the pair of linear motors **100** that apply the thrust in the X-axis direction and the thrust in the Y-axis direction to the stage body **60**. Each of the pair of linear motors **100** includes the first unit **200** having a plurality of the armature modules **210U**, **210V**, and **210W** each including: the magnetic core **201** having three protruding portions **201a**; and the coil **203** wound around the magnetic core **201** and through which currents of the same phase flow, and the second unit **300** having the magnet module **310** that includes a plurality of the permanent magnets **301** arranged while changing the poles in the X-axis direction and is accommodated between two adjacent protruding portions **201a**. At least a part of each of the permanent magnets **301** is accommodated in the space SP1 sandwiched between two adjacent protruding portions **201a** of the magnetic core **201**. Accordingly, since the magnetic attraction force MAF is generated between the first unit **200** and the second unit **300**, it is possible to control the force generated between the first unit **200** and the second unit **300** by controlling the d-axis current supplied to each of the pair of the linear motors **100**. As a result, the position of the stage body **60** in the Y-axis direction can be controlled by the pair of linear motors **100**, and it is not necessary to provide, for example, a voice coil motor or the like for position control of the stage body **60**. That is, the position of the stage body **60** in the Y-axis direction as well as the X-axis direction and the θ_z direction can be controlled with a simple configuration using the pair of linear motors **100**. Further, the linear motor **100** according to the present embodiment includes the magnetic core **201**,

which allows for a higher magnetic flux density than a coreless linear motor. This allows the stage body **60** to be driven at a higher speed than a coreless linear motor. As described above, in the mask stage device MST of the present embodiment, high-speed driving of the stage body **60** and position control of the stage body **60** in three directions (the X-axis direction, the Y-axis direction, and the θ_z direction) can be achieved with the pair of linear motors **100**.

[0082] Further, in the present embodiment, the stage body **60** is rotatable around the Z-axis direction orthogonal to the X-axis direction and the Y-axis direction. That is, the stage body **60** has three degrees of freedom. This makes it possible to synchronize the driving of the stage body **60** with the driving of the substrate stage.

[0083] Further, in the present embodiment, the pair of the linear motors **100** are opposed to each other in the Y-axis direction with the stage body **60** interposed therebetween. This allows the stage body **60** to be moved in both the +Y direction and the -Y direction.

[0084] Further, in the present embodiment, the direction of the thrust in the Y-axis direction applied to the stage body **60** by one of the pair of linear motors **100** is opposite to the direction of the thrust in the Y-axis direction applied to the stage body **60** by the other of the pair of linear motors **100**. This allows the stage body **60** to be moved in both the +Y direction and the -Y direction by controlling the magnitude relationship between the two thrust forces in opposite directions.

[0085] Further, in the present embodiment, the mask stage device MST includes the mask stage control device **400** that generates and controls the generation of the d-axis current and the q-axis current supplied to the coil **203**, and the mask stage control device **400** changes the thrust in the Y-axis direction applied to the stage body **60** by changing the d-axis current. This allows the position of the stage body **60** in the Y-axis direction to be changed.

[0086] Further, in the present embodiment, the mask stage control device **400** adjusts the thrust in the Y-axis direction applied to the stage body **60** by causing the d-axis current supplied to one of the pair of linear motors **100** to be different from the d-axis current supplied to the other of the pair of linear motors **100**. This allows the stage body **60** to be moved in both the +Y direction and the -Y direction.

[0087] Further, in the present embodiment, the mask stage control device **400** determines the d-axis current to be supplied to each of the pair of linear motors **100** based on the difference between the target position of the stage body **60** in the Y-axis direction and the actual position of the stage body **60** in the Y-axis direction. This allows the position of the stage body **60** to be brought closer to the target position in the Y-axis direction.

[0088] In the above embodiment above, the first unit **200** is fixed to the X beam **61**, and the second unit **300** is fixed to the stage body **60**, but this does not intend to suggest any limitation. FIG. **6A** is a plan view of a mask stage device MST-1 according to a variation, and FIG. **6B** is a side view of the mask stage device MST-1 according to the variation. FIG. **6B** illustrates a cross section including the permanent magnet **301** of the second unit **300**.

[0089] As illustrated in FIG. **6A** and FIG. **6B**, the first unit **200** may be fixed to the stage body **60**, and the second unit **300** may be fixed to the X beam **61**. In this case, the first unit **200** may include one armature module set **211** including the armature modules **210U**, **210V**, and **210W**, or may include a plurality of armature module sets. Other configurations are the same as those of the embodiment, and thus detailed description thereof will be omitted.

[0090] In the above embodiment, the magnetic core **201** of the armature module **210** of the first unit **200** has three protruding portions **201a**, and the second unit **300** has two magnet modules **310**. However, this does not intend to suggest any limitation. For example, the magnetic core **201** of the armature module **210** of the first unit **200** may have two protruding portions **201a**, and the second unit **300** may have one magnet module **310**. In this case, one magnet module **310** is disposed between two protruding portions **201a**. The magnetic core **201** of the armature module **210** of the first unit **200** may include N (N is a positive integer of 4 or greater) protruding portions **201a**, and the second unit **300** may include N-1 magnet modules **310**.

[0091] Further, in the above embodiment, the case has been described where the exposure device **10** is an exposure device that transfers the pattern of the mask MSK onto the glass substrate, but the exposure device **10** may be, for example, a semiconductor exposure device that forms a pattern formed on a reticle onto a wafer.

[0092] In addition to the exposure device **10**, the above embodiment may be applied to a device that holds an object and controls its position in two directions that intersect in the horizontal plane.

[0093] The above embodiments are preferred examples. However, the present disclosure is not limited to this, and various modifications can be made without departing from the scope of the present disclosure, and arbitrary constituent features may be combined.

Claims

1. An object holding device comprising: a holding unit that holds an object and is driven in a first direction, which is a scanning direction of the object, and a second direction orthogonal to the first direction in a horizontal plane; and a pair of linear motors that apply a thrust in the first direction and a thrust in the second direction to the holding unit, each of the pair of linear motors including a first unit and a second unit, the first unit including a plurality of armature modules, each of the armature modules including a magnetic core having two or more protruding portions, which protrude in the second direction, and a coil wound around the magnetic core, currents of the same phase flowing through the coil, and the second unit including a magnet module that includes a plurality of permanent magnets arranged in the first direction while changing poles and is arranged between two adjacent protruding portions of the two or more protruding portions, wherein at least a part of each of the permanent magnets is accommodated in a space sandwiched between the two adjacent protruding portions.
2. The object holding device according to claim 1, wherein another part of each of the permanent magnets, which is opposite to the protruding portions, is exposed from the space.
3. The object holding device according to claim 1, wherein the holding unit is rotatable around a third direction orthogonal to the first direction and the second direction.
4. The object holding device according to claim 1, wherein the pair of linear motors face each other in the second direction across the holding unit.
5. The object holding device according to claim 4, wherein a direction of a first thrust in the second direction applied to the holding unit by one of the pair of linear motors is opposite to a direction of a second thrust in the second direction applied to the holding unit by the other of the pair of linear motors.
6. The object holding device according to claim 1, further comprising: a current control unit configured to generate and control a d-axis current and a q-axis current to be supplied to the coil, wherein the current control unit changes the thrust in the second direction applied to the holding unit by changing the d-axis current.
7. The object holding device according to claim 6, wherein the current control unit adjusts the thrust in the second direction applied to the holding unit by causing a d-axis current supplied to one of the pair of linear motors to be different from a d-axis current supplied to the other of the pair of linear motors.
8. The object holding device according to claim 6, wherein the current control unit determines a d-axis current to be supplied to each of the pair of linear motors based on a difference between a target position of the holding unit in the second direction and an actual position of the holding unit in the second direction.
9. The object holding device according to claim 1, wherein the second unit is fixed to the holding unit.
10. The object holding device according to claim 1, wherein the first unit is fixed to the holding unit.

11. An exposure device comprising: the object holding device according to claim 1; and a pattern formation device that forms a pattern that the object has onto an exposure target by an exposure operation of exposing the exposure target with an energy beam through the object held by the object holding device.

12. An object moving method comprising: arranging a pair of first units so that a first direction is orthogonal to a scanning direction of an object, each of the pair of first units having a plurality of armature modules, each of the armature modules including a magnetic core having two or more protruding portions, which protrude in the first direction, and a coil wound around the magnetic core, currents of the same phase flowing through the coil; arranging a pair of second units each having a magnet module that includes a plurality of permanent magnets arranged in a second direction while changing poles in the second direction and is arranged between two adjacent protruding portions of the two or more protruding portions so that the second direction is parallel to the scanning direction and at least a part of each of the permanent magnets is accommodated in a space sandwiched between the two adjacent protruding portions; and applying a thrust in the scanning direction and a thrust in a direction orthogonal to the scanning direction in a horizontal plane to a holding unit that holds the object by a pair of linear motors each including the first unit and the second unit to move the object in the scanning direction and the direction orthogonal to the scanning direction.

13. An object holding system comprising: a holding unit that holds an object and is driven in a first direction, which is a scanning direction of the object, and a second direction orthogonal to the first direction in a horizontal plane; a pair of linear motors that apply a thrust in the first direction and a thrust in the second direction to the holding unit, each of the pair of linear motors including a first unit and a second unit, the first unit having a plurality of armature modules, each of the armature modules including a magnetic core having two or more protruding portions, which protrude in the second direction, and a coil wound around the magnetic core, currents of the same phase flowing through the coil, the second unit having a magnet module that includes a plurality of permanent magnets arranged in the first direction while changing poles and is arranged between two adjacent protruding portions of the two or more protruding portions; and a control device configured to control the pair of linear motors, wherein at least a part of each of the permanent magnets is accommodated in a space sandwiched between the two adjacent protruding portions.
